

Estimation of the Residual Error due to Ionospheric Bending in Relative TEC using COSMIC-2 GPS-RO Measurements

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BIOGRAPHY

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ABSTRACT

The Total Electron Content (TEC) derived from Global Positioning System (GPS) radio occultation (RO) measurements contains residual bending errors that arise from signal refraction and dispersion in the ionosphere. Motivated by the need to ensure the accuracy of TEC retrievals, particularly under high solar activity conditions, we investigate the impact of ionospheric bending on the conventional dual-frequency TEC. This work marks a new attempt in assessing the residual bending error in TEC estimates using real RO measurements, in contrast to prior research efforts that relied primarily on simulation studies. We present an estimation method for the residual bending error based solely on measurement data, avoiding reliance on a priori knowledge of the true ionospheric state. Utilizing real RO observations from COSMIC-2, our analysis reveals that the residual bending error is largely influenced by the vertical gradient of TEC, with additional dependencies on solar activity and local time. Large residual bending errors were observed under high solar activity conditions, reaching up to 17.7 TECU in a 15-day dataset from January 2023. Residual bending errors in dual-frequency TEC were often found to exceed the 3 TECU threshold from COSMIC-2 accuracy requirements, suggesting the use of single-frequency TEC or the application of residual bending error correction as a potential alternative under certain conditions inducing high TEC gradients.

1 INTRODUCTION

Radio signals experience delay and bending as they propagate through the ionosphere, with the degree of bending varying with their frequencies due to the dispersive nature of the ionosphere. In ionospheric Global Navigation Satellite System (GNSS) radio occultation (RO), the signals transmitted from GNSS satellites are received by Low Earth Orbit (LEO) satellites as they rise or set through the ionosphere. The collected measurements contain information about the slant Total Electron Content (TEC), representing the integrated electron density along the ray paths. The standard approach for retrieving TEC involves taking a linear combination of the phase measurements at two different frequencies, such as the L1 and L2 signals for the Global Positioning System (GPS) satellites, in order to extract the ionospheric term from the occultation measurements. This method effectively cancels out the geometric distance and clock biases, and is currently implemented in the TEC processing algorithm at COSMIC Data Analysis and Archive Center (CDAAC) (Yue et al., 2011). However, the key assumption in dual-frequency TEC estimation is straight-line

propagation, which assumes that the signals at different frequencies travel along the same straight-line path. Due to this neglect of ionospheric bending and dispersion, the estimated dual-frequency TEC deviates from the true TEC corresponding to the ideal straight-line path (Høeg et al., 1998). This is known as the residual error due to ionospheric bending, which we refer to simply as the residual bending error.

It is known that the residual bending error is dependent on the gradient of refractivity, which corresponds to the vertical gradient of the electron density to a first-order approximation in the context of ionospheric RO (Hajj & Romans, 1998). The magnitude of this error can vary considerably depending on the solar activity, local time and geographical location of the occultation event. Under nominal solar activity conditions, the residual bending error tends to be negligible, allowing the dual-frequency TEC estimate to be a close approximation of the true straight-line TEC (Schreiner et al., 1999). However, simulation studies have shown that the maximum deviation between the actual ray path and the straight line can reach up to a few kilometers at solar maximum conditions (Hajj & Romans, 1998; Hajj et al., 2000; Schreiner et al., 1999), resulting in residual bending errors on the order of 10 TECU (1 TECU = 10^{16} electrons/m²) (Høeg et al., 1998; Hoque & Jakowski, 2010). In such cases, the accuracy of dual-frequency TEC estimation can be significantly reduced and may not satisfy the mission requirement for the absolute TEC accuracy, which is 3 TECU for the Constellation Observing System for Meteorology, Ionosphere, and Climate-2 (COSMIC-2) mission (Yue et al., 2014). Hence, as the current solar cycle 25 approaches its maximum phase, there is a growing need for a comprehensive assessment of the residual bending error in TEC estimates from occultation measurements to apply appropriate error correction and enhance their accuracy.

The impact of higher-order ionospheric effects including ionospheric bending on neutral atmospheric retrievals has been studied extensively over the years (Danzer et al., 2013; Gorbunov et al., 1996; Healy & Culverwell, 2015; Mannucci et al., 2011), yet their influence on ionospheric retrievals remains relatively less explored. Høeg et al. (1998) and Syndergaard (2000) presented analytical models to characterize the residual bending error as a function of TEC, electron density, and the tangent altitude of the ray path under the assumption of spherical symmetry. Hoque & Jakowski (2011) proposed an empirical model for the residual bending error based on simulation results, which requires an additional ionospheric parameter for maximum ionization such as NmF2. Nevertheless, applying these models in practical cases presents a challenge as they require a priori knowledge of the electron density distribution. Although an approximate distribution of the electron density in the ionosphere can be reconstructed from the occultation measurements, the residual bending error calculated with these estimates generally deviates from the true residual bending error (Høeg et al., 1998; Syndergaard, 2002). This limitation has been acknowledged by Syndergaard (2002), who proposed a new method of combining the L1 and L2 excess phase data to eliminate the effects of bending and dispersion to the first order. Utilizing the proportionality of the frequency-dependent residual terms due to ray path bending in the L1 and L2 phase observables, as derived by Gu & Brunner (1990) and Syndergaard (2000), this approach allows for the mitigation of the residual bending error solely based on dual-frequency phase measurements and precise orbit determination (POD) data, without requiring knowledge of the true electron density distribution. Although validated through ray tracing with simple ionospheric models, this methodology has not undergone validation using real measurements.

Building on this approach, we propose an estimation method for the residual bending error based purely on measurements, without utilizing any information about the electron density distribution in the ionosphere. We conduct a comprehensive assessment of the residual bending error in dual-frequency TEC using GPS radio occultation measurements obtained from COSMIC-2, combining empirical analysis with a theoretical basis. This paper focuses on the bending error in relative TEC, derived exclusively from phase measurements and disregarding constant offsets due to phase ambiguities. The phase ambiguity can be removed by leveling the relative TEC to the unambiguous TEC obtained from pseudorange measurements. However, potential inaccuracies resulting from ionospheric bending effects in the pseudorange measurements are not within the scope of this paper.

The outline of this paper is as follows. The theory and mathematical derivations of the residual ionospheric error, accounting for the second-order ionospheric term as well as the bending error, are presented in Section 2 based on previous theoretical works. We present the proposed method for residual bending error estimation at the end of Section 2. Subsequently, the ionospheric excess phase processing required for our methodology is briefly reviewed in Section 3. Section 4 presents the results of the residual bending error analysis, encompassing case studies and statistical analyses, with conclusions following in Section 5.

2 DERIVATION OF THE RESIDUAL IONOSPHERIC ERROR

The mathematical derivation of the residual ionospheric error in TEC estimates, including the ray path bending term and the second-order term related to the geomagnetic field, is well described in Syndergaard (2002). The ideal RO phase measurement for

GPS signals propagating through the ionosphere can be derived by the line integral of the refractive index along the ray path between the GPS and LEO satellites:

$$L_i = \int_i n ds = \int_i \left(1 - C \frac{N_e}{f_i^2} \pm \frac{KB_{||}N_e}{f_i^3} \right) ds \quad (1)$$

where n is the refractive index given by the Appleton-Hartree formula with the series expansion to second order, f is the carrier frequency, N_e is the electron density, $B_{||}$ is the absolute value of the component of the geomagnetic field along the ray direction, $C = 40.3082 \text{ m}^3/\text{s}^2$, and $K = 1.1283 \times 10^{12} \text{ m}^3/\text{T s}^3$. The integer ambiguities, transmitter and receiver clock biases, relativistic effects and random errors are disregarded. The subscript i represents the signal frequency ($i = 1, 2$) and indicates that the signals are propagating through curved ray paths. This gives rise to a residual error when each term is rewritten with its difference from the integrals taken over the straight-line path as denoted by the subscript 0.

$$L_i = \int_0 ds - \frac{C}{f_i^2} \int_0 N_e ds \pm \frac{K}{f_i^3} \int_0 B_{||} N_e ds + \left(\int_i ds - \int_0 ds \right) - \frac{C}{f_i^2} \left(\int_i N_e ds - \int_0 N_e ds \right) \quad (2)$$

The last two terms in brackets represent the error terms due to ray path bending. Using a series expansion, these bending terms were shown to be mainly proportional to f_i^{-4} (Gu & Brunner, 1990; Syndergaard, 2000). Because the term related to the geomagnetic field (third term in Equation (2)) is small enough compared to the bending term, it can be represented by its straight-line approximation (Syndergaard, 2002). Hence, Equation (2) is rewritten as

$$L_i = \int_0 ds - \frac{C}{f_i^2} \int_0 N_e ds \pm \frac{K}{f_i^3} \int_0 B_{||} N_e ds - \frac{1}{2} \frac{C^2}{f_i^4} \Gamma \quad (3)$$

where Γ is dependent on the square of the vertical gradient of the TEC and the vertical gradient of the integral of the squared electron density, as expressed in Equation (4). This expression was derived under the assumption of spherical symmetry and zero electron density at and above the LEO orbit altitude (Høeg et al., 1998; Syndergaard 2000).

$$\Gamma = D \zeta \left(\frac{d}{da} \int_0 N_e ds \right)^2 + a \frac{d}{da} \int_0 N_e^2 ds \quad (4)$$

Here, a is the straight-line tangent altitude, and D is the reduced distance, defined as

$$\frac{1}{D} = \frac{1}{\sqrt{r_L^2 - a^2}} + \frac{1}{\sqrt{r_G^2 - a^2}} \quad (5)$$

where r_L and r_G are LEO and GPS orbit radial distances. ζ is the defocusing factor defined as

$$\zeta = \left(1 - D \frac{d\alpha}{da} \right)^{-1} \quad (6)$$

where α is the bending angle of the ionosphere-free path. For ionospheric altitudes with negligible influence from the neutral atmosphere, the defocusing factor is taken to be unity (Høeg et al., 1998).

The traditional dual-frequency method for TEC estimation uses the L1-L2 linear combination of phase measurements, as follows:

$$TEC_{12} = \frac{f_1^2 f_2^2 (L_1 - L_2)}{C(f_1^2 - f_2^2)} \quad (7)$$

Substituting Equation (3) into (7), we obtain

$$TEC_{12} = TEC_0 + \Delta TEC_{12}^{2nd} + \Delta TEC_{12}^{bend} \quad (8)$$

where

$$TEC_0 = \int_0 N_e ds \quad (9)$$

$$\Delta TEC_{12}^{2nd} = \mp \frac{K(f_1^3 - f_2^3)}{C f_1 f_2 (f_1^2 - f_2^2)} \int_0 B_{||} N_e ds \quad (10)$$

$$\Delta TEC_{12}^{bend} = \frac{C}{2} \frac{f_1^2 + f_2^2}{f_1^2 f_2^2} \Gamma \quad (11)$$

TEC_0 is the straight-line TEC, which is the true value to be estimated. ΔTEC_{12}^{2nd} is the error due to the geomagnetic field effect, and ΔTEC_{12}^{bend} is the error due to the ionospheric bending in the dual frequency-based TEC.

If the POD information for the GPS and LEO satellites are available, TEC can also be determined using only the L1 frequency measurements, as given by:

$$TEC_1 = -\frac{f_1^2}{C} S_1 \quad (12)$$

where S_1 is the ionospheric delay contributing to the L1 phase measurement, referred to as the L1 ionospheric excess phase. For the ideal phase measurement given in Equation (1), the ionospheric excess phase is expressed as

$$S_1 = L_1 - \int_0 ds \quad (13)$$

Then, substituting Equation (3) into (12) and (13), we obtain

$$TEC_1 = TEC_0 + \Delta TEC_1^{2nd} + \Delta TEC_1^{bend} \quad (14)$$

where

$$\Delta TEC_1^{2nd} = \mp \frac{K}{C f_1} \int_0 B_{||} N_e ds \quad (15)$$

$$\Delta TEC_1^{bend} = \frac{C}{2} \frac{1}{f_1^2} \Gamma \quad (16)$$

Equations (15) and (16) are residual errors from the geomagnetic field and the bending term in the single-frequency TEC, respectively, corresponding to Equations (10) and (11) in the dual-frequency TEC.

Equations (11) and (16) demonstrate a proportional relationship between the residual bending errors in the dual- and single-frequency TEC, which can be expressed as a function of signal frequency. Syndergaard (2002) derived and validated this result through a ray tracing simulation, and proposed a new algorithm for TEC estimation that utilizes the combination of L1 and L2 excess phases to cancel out the bending term. Similar to this approach, we may estimate the bending term Γ by eliminating the straight-line TEC from Equations (8) and (14):

$$\Gamma = \frac{2f_2^2}{C} (TEC_{12} - TEC_1) \pm \frac{2K}{C^2} \frac{f_1 f_2}{f_1 + f_2} \int_0 B_{||} N_e ds \quad (17)$$

It should be noted that the same result can be obtained by taking the ionosphere-free combination of the phase measurements in Equation (3) to eliminate the first-order ionosphere term and subtracting the geometric distance:

$$\Gamma = \frac{2f_1^2 f_2^2}{C^2} \left(\frac{f_1^2 L_1 - f_2^2 L_2}{f_1^2 - f_2^2} - \int_0 ds \right) \pm \frac{2K}{C^2} \frac{f_1 f_2}{f_1 + f_2} \int_0 B_{||} N_e ds \quad (18)$$

The influence of the geomagnetic field term on the residual ionosphere error is notably smaller compared to the bending term, typically accounting for less than 0.1% of the TEC value, as indicated by Høeg et al. (1998). It is often excluded from initial derivations due to its negligible impact, but can nonetheless be handled separately using models of the geomagnetic field and ionospheric electron density. Thus, the geomagnetic field error is omitted for simplicity in the following, and its rough estimate using real measurements will be provided in Section 4. Then, substituting Equation (17) to (11), the residual bending error in the dual-frequency TEC estimate is given as

$$\Delta TEC_{12}^{bend} \cong \frac{f_1^2 + f_2^2}{f_1^2} (TEC_{12} - TEC_1) \quad (19)$$

Similarly, the bending error in the single-frequency TEC estimate can be calculated by substituting Equation (17) into (16). This method enables a reliable estimation of the residual bending error in the TEC estimates, calculated purely from real measurements without requiring knowledge of the true ionosphere profile.

3 IONOSPHERIC EXCESS PHASE PROCESSING

As briefly mentioned in Section 2, the single-frequency TEC estimation requires the extraction of the ionospheric excess phase from the L1 frequency phase measurement. This process is similar to the atmospheric excess phase processing required for the calculation of bending angles in the neutral atmosphere, which is detailed in Schreiner et al. (2010). While the expression for the ionospheric excess phase in Equation (13) is for an ideal phase measurement, the actual phase measurement includes many more terms which need to be estimated and removed in order to isolate the ionospheric excess phase (Schreiner et al., 2010):

$$\hat{L}_i = \rho_r^s + \delta \rho_{r,rel}^s + c \cdot (\delta t_r + \delta t_{r,rel}) - c \cdot (\delta t^s + \delta t_{rel}^s) - \delta \rho_{i,iono} + \lambda_i N_i + \epsilon_i \quad (20)$$

$\hat{L}_i$ denotes the observed phase measurement in units of length for a signal with wavelength λ_i , $\rho_r^s (= \int_0 ds)$ is the geometric distance between the GPS (satellite s) and LEO (receiver r) satellites, δt_r is the LEO clock bias at the reception time, δt^s is the GPS clock bias at transmission time, $\delta t_{r,rel}$ and δt_{rel}^s are the relativistic effects on the LEO and GPS clocks respectively, $\delta \rho_{r,rel}^s$ is the relativistic gravitational delay, N_i is the integer ambiguity, and ϵ_i is the phase noise. $\delta \rho_{i,iono}$ is the frequency-dependent ionospheric delay, which can be expressed as Equation (21) including the residual error from the geomagnetic field and ray path bending:

$$\delta \rho_{i,iono} = \frac{C}{f_i^2} \int_0 N_e ds \mp \frac{K}{f_i^3} \int_0 B_{||} N_e ds + \frac{1}{2} \frac{C^2}{f_i^4} \Gamma \quad (21)$$

Here, we restrict our region of interest to altitudes above approximately 80 km, as we are only concerned with the ionosphere within the context of this paper. The measurements collected within this altitude range will contain negligible neutral atmospheric contribution to the signal delay and bending, thus these effects are omitted in Equation (20). Neglecting phase noise and the effect from the integer ambiguity, the L1 ionospheric excess phase can be determined up to a constant bias by removing all terms in Equation (20) apart from the ionospheric delay.

$$\hat{S}_1 = -\delta \rho_{1,iono}$$

$$= \hat{L}_1 - \{\rho_r^s + \delta\rho_{r,rel}^s + c \cdot (\delta t_r + \delta t_{r,rel}) - c \cdot (\delta t^s + \delta t_{rel}^s)\} \quad (22)$$

Substituting Equation (22) into (12) will yield the single-frequency TEC estimate.

The following subsections outline key aspects of ionospheric excess phase processing that have a significant impact on the accuracy of the retrieved excess phase. These include the removal of the GPS and LEO clock biases and the geometric distance from the phase measurement.

3.1 Removal of GPS and LEO clock biases

For the accurate retrieval of the ionospheric excess phase, it is crucial to select a suitable method for removing GPS and LEO satellite clock biases. In this study, we employ a single-difference processing strategy that effectively eliminates the LEO clock bias by differencing data from the occulting GPS satellite with simultaneously sampled data from a reference GPS satellite (Schreiner et al., 2010). GPS satellite clock errors are corrected using prior estimated solutions. This approach outperforms zero-difference (or un-difference) processing, which relies on prior estimates for both LEO and GPS clocks and is only practical for LEO clocks with ultra-stable oscillators (Beyerle et al., 2005). Since the available COSMIC-2 POD data from CDAAC is currently sampled at 1-minute intervals, it cannot adequately capture the higher frequency variations of the LEO clock, making zero-differencing impractical for COSMIC-2. Single-differencing is also preferred over the double-differencing procedure, which involves ground receiver measurements prone to errors such as multipath and neutral atmospheric effects (Schreiner et al., 2010). Therefore, we adopt single-differencing with a high-rate GPS clock solution that sufficiently captures the clock fluctuations. We utilize GPS clock corrections at 5-second sampling, provided by Center of Orbit Determination in Europe (CODE) (Bock et al., 2009).

The selection of the reference GPS satellite for single-differencing is based on the criteria that it must be tracked simultaneously with the occulting GPS satellite throughout the occultation event and have a positive elevation above the local horizon of the LEO satellite. The ionosphere-free observation from this reference link will contain only the terms related to the LEO clock, once the reference GPS clock bias and geometric distance are removed. By utilizing reference GPS observations at positive elevations, any residual ionosphere error left after taking the ionosphere-free combination of the reference link observations will be negligible. Subtracting the ionosphere-free measurement of the reference link from the occulting link measurement completes the removal of the LEO clock bias.

3.2 Removal of geometric distance

Accurate determination and removal of the geometric distance rely on precise orbit data for both LEO and GPS satellites, referenced to their mass centers. Since the true geometric distance is between the antenna phase centers of the transmitting and receiving antennas, adjustments are necessary to account for the antenna offset and the orientation of the offset vector in space (Schreiner et al., 2010). While the GPS satellite maintains a relatively stable orientation during occultation events, rendering its antenna offset and attitude effects negligible for RO processing (Hunt et al., 2018; Innerkofler et al., 2023), these effects must be considered for the LEO satellite. In this work, we implement an antenna offset correction for the LEO satellite by accounting for the vector from the center-of-mass to the antenna reference point (ARP). The coordinate transformation of the ARP vector from the satellite body frame to Earth-Centered Inertial (ECI) coordinates utilizes quaternions in the COSMIC-2 attitude files provided by CDAAC, accounting for changes in the LEO satellite orientation (Zhang et al., 2022). The ARP vector in ECI is then added to the mass center position to retrieve the precise position estimates of the LEO satellite antenna. The phase center offset (PCO) and phase center variation (PCV) are not considered in this paper, but will be explored in future studies as they are needed to enhance the precision of the calculated geometric distance.

4 RESULTS AND DISCUSSION

In this study, we evaluate the residual bending error in TEC estimates using measurements collected from the POD antenna of COSMIC-2 flight model (FM) 6. COSMIC-2 satellites have two POD antennas, front and rear, which are mounted at a tilted angle relative to the zenith direction (Weiss et al., 2020). This configuration allows sufficient coverage of the ionospheric region both above and below the LEO orbit altitude, corresponding to positive and negative elevations respectively. The POD antenna

observations are used for satellite orbit and clock determination as well as ionospheric retrievals. Additional data products required for the ionospheric excess phase processing as described in Section 3 are given in Table 1.

TABLE 1

Data Products used for Single- and Dual-frequency TEC Estimation and Residual Bending Error Assessment

Type (Sampling interval)	File Format	Source
POD antenna observations (1 sec)	podCrx (RINEX)	COSMIC Data Analysis and Archive Center (CDAAC)
LEO satellite attitude (1 sec)	leoAtt	
LEO precise orbit and clock (1 min)	leoOrb (SP3)	
GPS final orbit (5 min)	*.SP3	Center of Orbit Determination in Europe (CODE)
GPS high-rate clock (5 sec)	*.CLK	
Earth orientation parameters (12 hr)	*.ERP	

The analysis utilized a dataset spanning 15 consecutive days for each period of low and high solar activity: July 28–August 11, 2020 and January 3–17, 2023, respectively. Profiles lacking observations of the lower ionosphere below LEO orbit altitude, as well as those exhibiting outliers, cycle slips, or other data quality issues such as inaccuracies in prior estimated orbit or clock solutions were excluded in the analysis.

4.1 Case Study

Two occultation events from different solar activity conditions were selected to examine ionospheric bending effects in TEC estimates using radio occultation signals. Case 1 is an occultation event that occurred on July 28, 2020 at 13:02–13:19 Universal Time (UT) (Figure 1), while Case 2 took place on January 13, 2023 at 04:38–04:50 UT (Figure 2). In both Figures 1 and 2, plot (a) illustrates the single and dual-frequency TEC estimates in red and blue, respectively, plotted against the elevation angle. Both cases represent setting occultation events, in which the signal to the LEO satellite transitions from traveling through the topside ionosphere to passing below the LEO orbit through the bottomside ionosphere, as the elevation angle shifts from positive to negative. Consequently, TEC values are lower at positive elevation angles, reflecting only the topside ionosphere contribution to signal delay, and increase as the elevation angle becomes negative. Note that the TEC estimates shown in this paper are relative values, where they were arbitrarily set to zero at the start of the profile. This neglects the topside ionosphere and the plasmasphere contribution to the TEC values, which may be appreciably large in high solar activity conditions and should be addressed in a future study. Towards the end of the occultation event when the elevation angle drops below -20° , the single-frequency TEC estimate diverges from the dual-frequency TEC. This is because the single-frequency TEC estimates are corrupted by neutral atmospheric delay as the signal traverses neutral atmospheric altitudes, whereas this effect is canceled out in dual-frequency estimates.

Plot (b) displays the residual bending error computed using Equation (19), with the curve in gray showing the invalid portion introduced from the neutral atmosphere. It is evident from both Figures 1 and 2 that residual bending error in the topside ionosphere is negligible, consistent with its relatively low electron density and TEC gradient. However, upon comparing the negative elevation portion, Figure 2(b) exhibits a large error peak, while Figure 1(b) does not show a notable difference in bending errors between the topside and bottomside ionosphere. This observation is supported by plot (c), which shows the tangent point trajectory of the COSMIC-2 satellite in white, with the larger dot indicating the location of the tangent point corresponding to the maximum residual bending error. The tangent point is defined for negative elevation observations traversing the ionosphere below orbit altitude. The vertical TEC maps were derived from the ground-based International GNSS Service (IGS) ionosphere products provided in ionosphere exchange (IONEX) format. In Figure 1(c), the ray tangent point travels from 18.7°S , 63.1°W to 7.9°S , 50.9°W , with the local time (LT) at maximum bending error being 9:54 and the vertical TEC around 14 TECU. In Figure 2(c), the ray tangent point travels from 16.4°N , 150.9°W to 12.4°N , 150.8°W , closer to the equatorial ionization anomaly (EIA) region that exhibits a significant spatial gradient of electron density. The maximum bending error occurs at 18:45 LT where the vertical TEC is approximately 53 TECU. From the ray path geometry, we observe that Case 2 experiences higher refractivity gradients and thus a larger ray path bending.

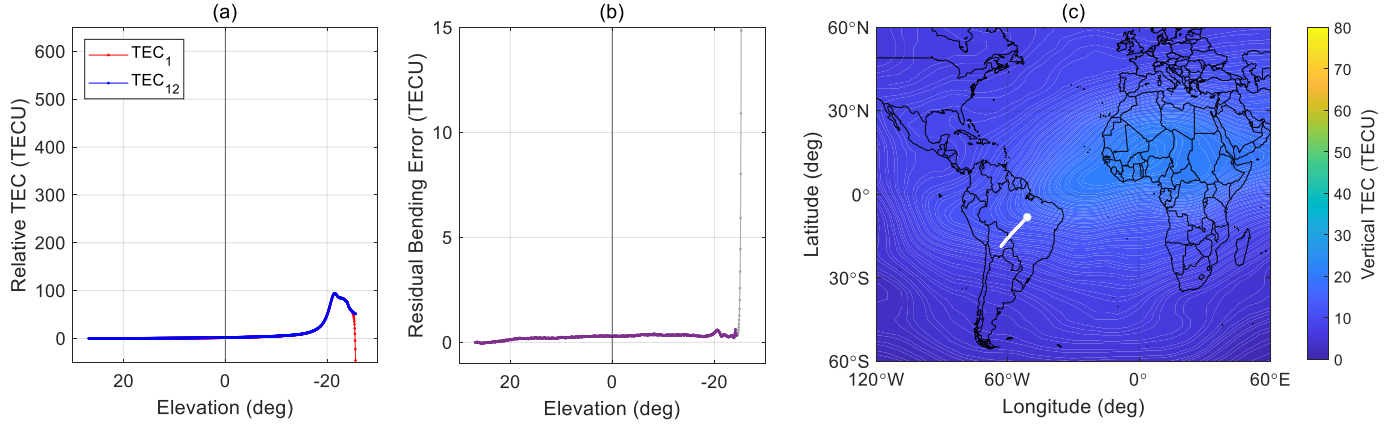


FIGURE 1 Residual bending error analysis for Case 1 (COSMIC-2 FM6 POD1 PRN17 2020/07/28 13:02–13:19 UT). (a) Single- (red) and dual-frequency (blue) TEC estimates. (b) Residual bending error in dual-frequency TEC. (c) Vertical TEC map and tangent point trajectory (white).

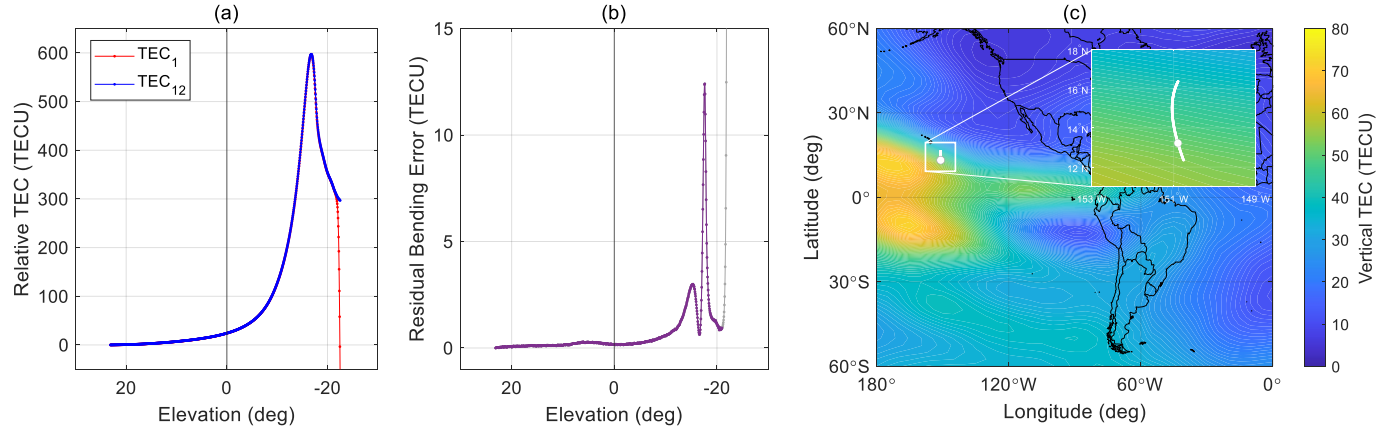


FIGURE 2 Similar to Figure 1, but for Case 2 (COSMIC-2 FM6 POD1 PRN22 2023/01/13 04:38–04:50 UT).

Figures 3 and 4 depict the relative TEC (plot (a)) and the residual bending error (plot (b)) as a function of altitude, taking only the negative elevation portion from Figures 1 and 2. The data is truncated at the altitude of 80 km to exclude the effect of the neutral atmosphere. Once again, these represent relative TEC values, but this time the initial point at the orbit altitude was set to zero. This signifies the ionosphere's extent only up to the height of the LEO orbit, above which no electron density exists, as assumed in the derivation of the bending term Γ alongside the spherical symmetry assumption (Syndergaard, 2000). However, the electron density at the orbit altitude might be considerable, particularly for a satellite in a relatively low orbit. COSMIC-2 FM6 was at an initial orbit altitude of 720 km in 2020 and was later lowered to the 530 km in 2021 (Weiss et al., 2022). Hence, the relative TEC in Figure 3(a) may closely approximate the absolute TEC value, given the negligible plasmasphere contribution especially under low solar activity conditions. Conversely, the relative TEC in Figure 4(a) is likely to contain appreciable errors from the topside ionosphere above 530 km, especially since Case 2 occurred during higher solar activity conditions. The impact of disregarding the topside ionosphere and plasmasphere contribution on residual bending error estimation will be addressed in future works.

The purple line in plot (b) represents the residual bending error. In Figure 4(b), this error extends up to 12.3 TECU when the vertical gradient is at its maximum, occurring at an altitude of around 200 km just below the F region peak. On the contrary, Figure 3(b) shows overall a significantly smaller bending error, with a maximum value of 0.32 TECU at an altitude of around 90 km, which is influenced by the electron density gradient at the pronounced E layer. The green dashed line is the residual bending error derived from the major dispersion term (first term in Equation (4)), $D(dTEC_0/da)^2$, where the unknown TEC_0 is replaced with the estimated TEC_{12} . The purple and green lines demonstrate a similar trend, confirming the main proportionality of the residual bending error to

the squared vertical gradient of TEC. To conduct a more comprehensive comparison between our measurement-based bending error estimation method and the theoretically derived bending error expression in Equation (4), we should also incorporate the second term in Equation (4), known as the minor dispersion term. This necessitates the knowledge of the electron density distribution to calculate the integral of N_e^2 , which can be done in practice by deriving the electron density profile from TEC_{12} using Abel inversion under the assumption of spherical symmetry (Høeg et al., 1998). However, it is important to note that this method may yield inaccurate results, particularly below the F layer, where the spherical symmetry assumption often introduces significant errors and may even yield unphysical, negative electron densities (Høeg et al., 1998; Syndergaard, 2006).

The yellow line in plot (b) represents a rough estimate of the error from the geomagnetic field in Equation (10). We have approximated the term $B_{||}$ with the total magnetic field intensity at the tangent point, assuming a ray path propagation direction parallel to the geomagnetic field. This provides a conservative estimation of the near worst-case contribution from the geomagnetic field. In Figure 3(b), $B_{||}$ has been set to 22.3 μT , which has been calculated for the magnetic field at 15°S latitude, 60°W longitude, and 80 km altitude with the use of International Geomagnetic Reference Field (IGRF) model. In Figure 4(b), $B_{||}$ has been set to 32.2 μT , for the magnetic field at 15°N latitude, 150°W longitude, and 80 km altitude. The results show that the maximum error from the geomagnetic term is 0.063 TECU and 0.56 TECU for Case 1 and 2 respectively, and can therefore be considered as having no appreciable impact on the residual ionospheric error. This justifies the omission of the geomagnetic term in Equation (17) for the derivation of the residual bending error in Equation (19).

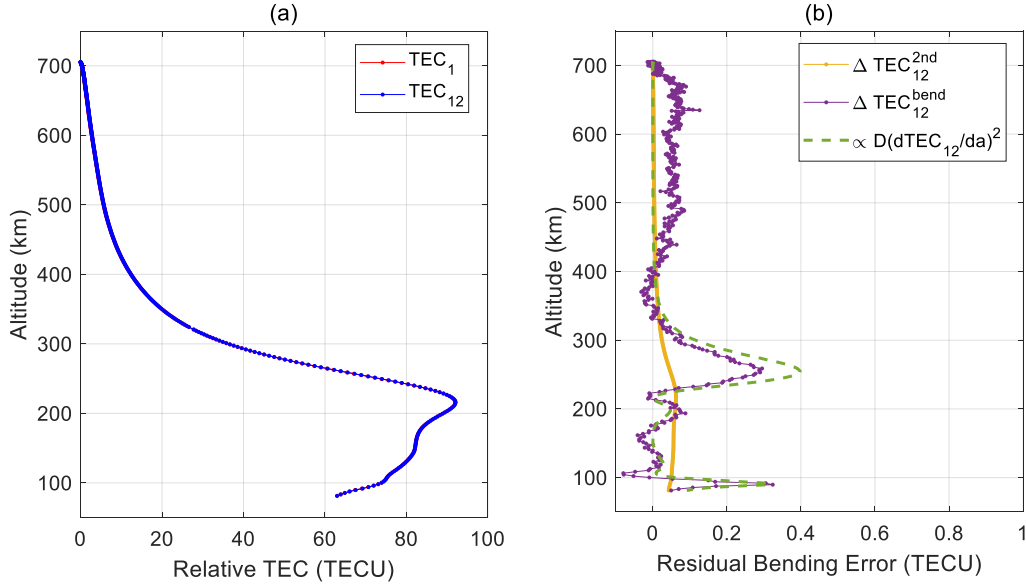


FIGURE 3 (a) Single- (red) and dual-frequency (blue) TEC estimates, (b) residual bending error (purple) and geomagnetic field error (yellow) as a function of altitude, for Case 1. The green dashed line is the residual bending error calculated with the first term (major dispersion term) in Equation (4).

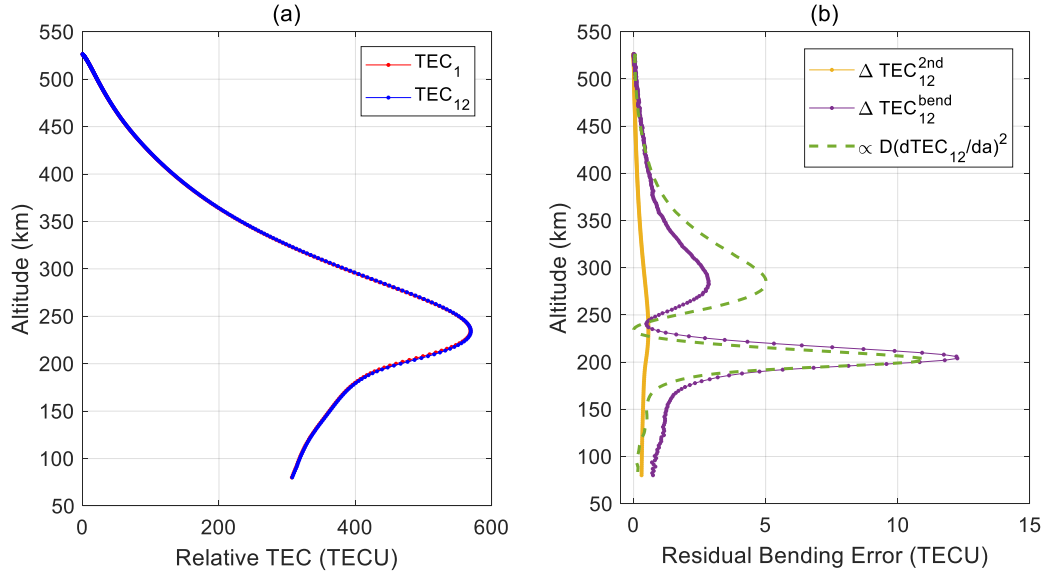


FIGURE 4 Similar to Figure 3, but for Case 2.

4.2 Statistical Study

The following subsection presents a statistical study assessing the correlation of the residual bending error with solar activity, local time, and vertical gradient of TEC. In Figure 5, the box plots display daily variations in the maximum residual bending error per event. Each box represents the interquartile range, with the lower and upper edges indicating the 25th and 75th percentiles, respectively. The median value is represented by a red line within each box. The data from 2023 generally exhibit a more dispersed distribution of the maximum residual bending errors compared to 2020, as indicated by a larger interquartile range and greater maximum values. This trend corresponds to an increased F10.7 solar flux index during periods of heightened solar activity, indicating that the residual bending error becomes more pronounced under higher solar activity conditions.

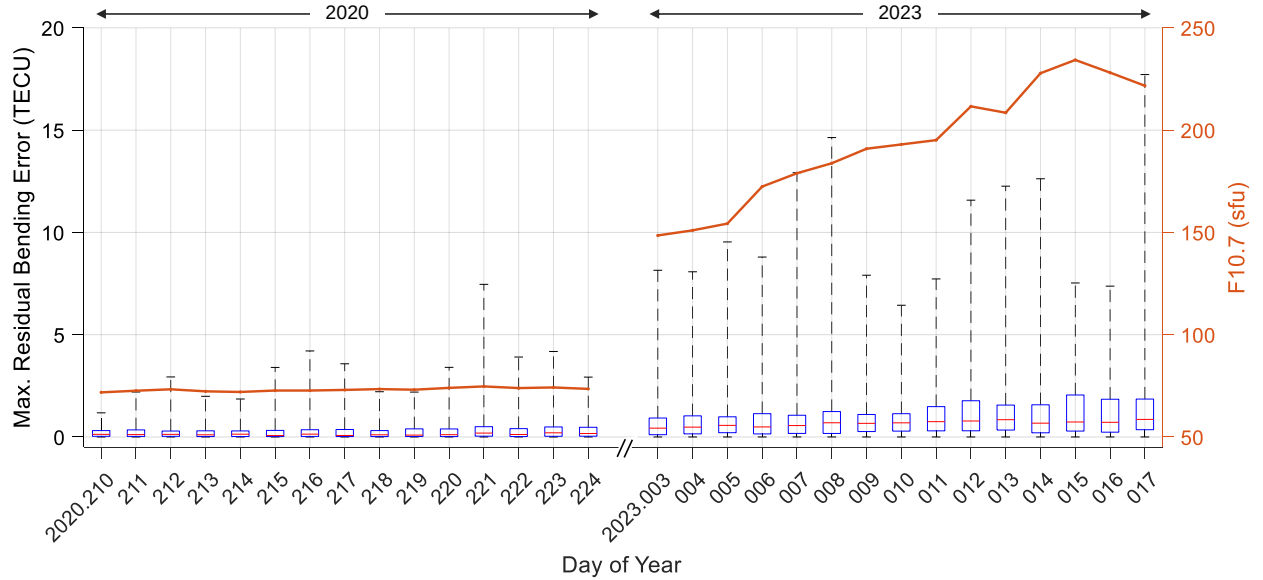


FIGURE 5 Daily variations of the maximum residual bending error distribution (left axis) and F10.7 solar flux index (right axis)

Here, we also observe that dual-frequency TEC estimates often contain residual bending errors that do not meet the 3 TECU accuracy requirement for COSMIC-2 absolute TEC products, especially during periods of high solar activity. In the dataset from 2020, 15 out of 2373 profiles (0.63%) exceeded the threshold, while in 2023, a much higher percentage of 232 out of 2389 profiles (9.71%) did so. It is worth noting that for single-frequency TEC, whose residual bending error is approximately 37.8% of that in dual-frequency TEC as obtained by comparing Equations (11) and (16), the number of profiles with residual bending errors higher than 3 TECU was only 25 out of 2389 (1.05 %) even in 2023.

Figure 6 presents a plot of the maximum residual bending error against the local time at the corresponding tangent point. It is observed that large residual bending errors often occur in the late afternoon, around 16:00-20:00 LT. This pattern may be related to the sharp spatial gradient of electron density around the solar terminator and the edge of the EIA, which becomes stronger during periods of high solar activity (Balan et al., 2018). Further analysis is required for a comprehensive understanding of the physical processes behind this observation.

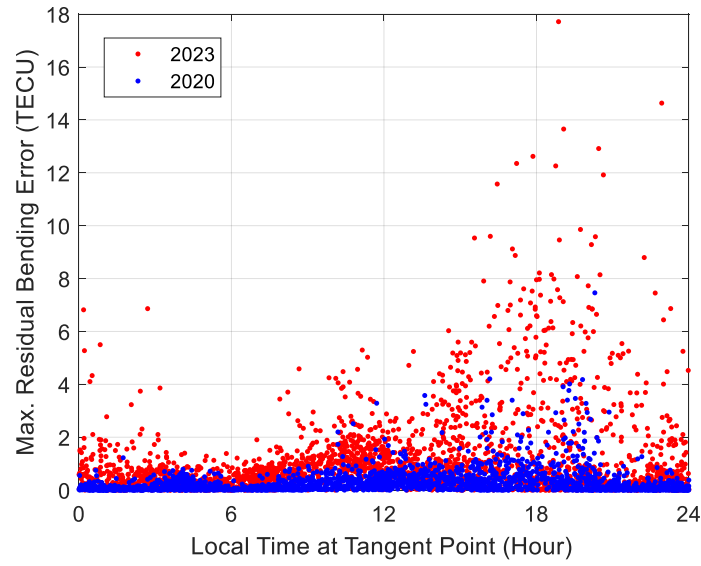


FIGURE 6 Scatter plot of maximum residual bending error against local time at corresponding tangent point

In Figure 7, we present a correlation study between the maximum residual bending error and the corresponding squared vertical gradient of the TEC. The plot displays a positive correlation, highlighting that the residual bending error is predominantly influenced by the TEC gradient. This validates that the residual bending error is larger at regions of high vertical TEC gradients corresponding to larger ray separations, which is consistent with the theoretical derivation by Syndergaard (2000). It should be noted that the relationship between residual bending errors and squared TEC gradients is not linearly proportional, as shown in Equation (4) where the expression for Γ is dependent on the occultation geometry and the vertical gradient of the integral of the squared electron density. Additionally, for cases where the residual bending error is less than about 1 TECU, it is difficult to distinguish actual ionosphere bending-related errors from other sources, such as POD errors. For instance, the computed residual bending error may be influenced by a drift in the single-frequency TEC due to velocity errors in the POD solution, as well as other data uncertainties such as those in the clock solutions.

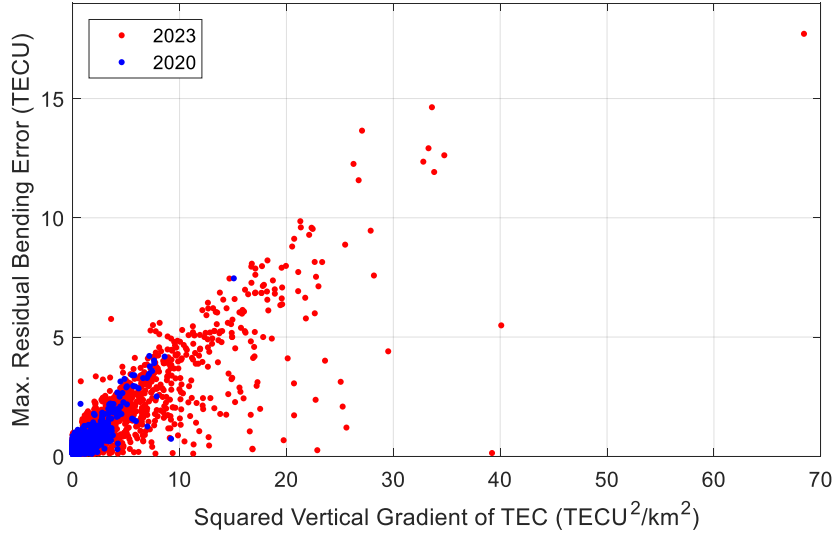


FIGURE 7 Scatter plot of maximum residual bending error against the corresponding squared vertical gradient of TEC

5 CONCLUSION

This study has presented a practical approach for assessing the residual bending error in the conventionally used dual-frequency TEC using real RO observations. Our approach estimates the residual bending error by utilizing the difference between the dual- and single-frequency TEC estimates, thereby deriving the bending error directly from measurements without the need of a priori information on the electron density distribution. Through an empirical analysis using real occultation measurements from COSMIC-2, we have demonstrated that residual bending errors can be substantial under high solar activity conditions, reaching up to 17.7 TECU within the 15-day dataset from January 2023. Additionally, statistical analyses indicated a significant dependency of the residual bending error on local time and vertical TEC gradient. The results demonstrated a strong correlation between residual bending errors and squared vertical TEC gradients, consistent with theoretical derivations (Syndergaard, 2000). As a rule of thumb, for vertical TEC gradients exceeding approximately 3 TECU/km (corresponding to the squared gradient of 9 TECU²/km²), there is a high probability of bending errors exceeding the 3 TECU threshold, which is the accuracy requirement of COSMIC-2 absolute TEC products (Yue et al., 2014).

We have observed that dual-frequency TEC estimates often contain residual bending errors that do not meet the 3 TECU accuracy requirement, especially for high solar activity periods. In the dataset from 2020, only a small fraction of profiles (less than 1%) exceeded this threshold, while in 2023, a significantly higher percentage (nearly 10%) did so. For single-frequency TEC, the residual bending errors are much lower compared to those in the dual-frequency TEC, with bending errors higher than 3 TECU being minimal (around 1%) even in 2023. This suggests that single-frequency TEC could serve as an alternative for dual-frequency TEC under certain conditions inducing high TEC gradients. Alternatively, the implementation of residual bending error correction, as proposed by Syndergaard (2002), may enhance the accuracy of the dual-frequency TEC estimates. Further analysis is necessary to establish specific thresholds or conditions necessitating the retrieval of single-frequency TEC or the correction of residual bending errors in dual-frequency TEC.

It is again emphasized that the discussion has been limited to the relative TEC within the scope of this paper. However, the bending error in the relative TEC will propagate to the absolute TEC, which is why we conservatively compare our estimated residual bending error with the accuracy requirement for the absolute TEC. In reality, the residual bending error in absolute TEC will increase from that in the relative TEC, since we estimate the constant bias from phase ambiguities using pseudorange measurements which are also influenced by ionospheric bending (Høeg et al., 1998; Syndergaard, 2002). The influence of ionospheric bending on phase-to-pseudorange leveling for absolute TEC retrieval from real RO measurements is the subject of our ongoing work. Additionally, as our work is built on the theoretical basis that is only strictly valid under the assumption of a spherically symmetric ionosphere and the neglect of the topside ionosphere and plasmasphere contributions, further analysis is needed to evaluate the effects of these simplifications, which may be the subject of our future efforts.

ACKNOWLEDGMENTS

Jaehee Chang was supported by the National Research Foundation of Korea funded by the Ministry of Science and ICT under Grant 2022M1A3C2069728, Future Space Education Center. The authors thank Dr. Stig Syndergaard of the Danish Meteorological Institute for helpful discussions and comments which have improved this paper.

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